

Mayank Shukla

✉ mayankshukla.ms.st@gmail.com | +918177063899 | 🏠 Website | 🐙 Github | 🔗 LinkedIn

Summary

- Multidisciplinary Computer Science and Electronics Engineer driven by a passion for devising real-world innovations through the cohesive integration of embedded hardware and advanced software systems.
- Demonstrates a comprehensive command over both hardware architecture and software engineering, enabling a systems-level perspective and cross-functional integration in complex product development cycles.
- Skilled in designing and implementing robust embedded platforms encompassing microcontrollers, sensor networks, real-time operating systems, and secure communication protocols for scalable, high-reliability applications.
- Committed to technological excellence and continuous innovation, leveraging emerging trends in IoT, AI ML, and cyber-physical systems to architect transformative and future-ready solutions.

Work Experience

Zuree Telecom Pvt Ltd

IoT Product Engineer

Gurugram, India

Oct. 2024 - Present

- Spearheading the end-to-end development of commercial IoT products, including system architecture, embedded firmware (C/C++/Python), and custom **PCB design** using platforms such as **ESP32, STM32, Raspberry Pi**, and **AMB82 MINI**.
- Developing ML model for various IoT Products.
- Integrated a wide range of sensors and modules—**SIM800L/A7670E (GSM)**, **MPU6050 (IMU)**, **GPS (NEO-6M/8M)**, **TTL converters**, **RS485 (MODBUS)**—with robust communication stacks using **MQTT, HTTP, CoAP and HTTPS** protocols for secure, cloud-connected applications.
- Contributed **two granted patents to the company**, focused on innovative IoT system architectures and sensor integration frameworks, reinforcing the organization's R&D and IP portfolio.
- Designed and deployed a custom-built quadcopter drone with embedded flight control, sensor fusion, and telemetry, demonstrating advanced control systems and real-time data acquisition capabilities.

TechVein IT Solution Pvt Ltd

Embedded Software Engineer

Kolkata, India

Jan. 2024 - Oct. 2024

- Developed and prototyped multiple IoT-based automation solutions, including line follower robots, pick-and-place robotic arms, and sensor-driven systems for industrial and educational use-cases.
- Conducted research and implementation of embedded control algorithms and real-time logic using microcontrollers (**ATmega, ESP8266, STM32**), integrated with IoT-compatible sensors for actuation and feedback loops.
- Designed systems utilizing a range of IoT communication protocols such as **MQTT, CoAP, HTTP**, and custom serial communication interfaces, ensuring reliable device-to-cloud connectivity and control.
- Evaluated and integrated various analog/digital sensors—proximity, infrared, temperature, ultrasonic, and encoders—for robotics and IoT applications, enabling real-time decision-making and adaptive behavior.

Education

BBDNITM

Bachelor of Technology, Computer Science

Thesis: [Anomaly Detection in Wireless Sensor Network](#).

Lucknow, UP, India

Sep. 2021 – July. 2024

- Focused on data structures, algorithms, operating systems, computer networks, and software engineering, with practical applications in **IoT and embedded systems**.
- Completed academic and capstone projects involving machine learning, IoT system design, cloud integration, and real-time data processing.
- Gained proficiency in programming languages including **C, C++, Python**, and **Java**; developed full-stack applications and microservices architectures.
- Applied principles of software design patterns, database systems, and version control in collaborative development environments using **Git and Agile methodologies**.

- Specialized in **analog and digital circuit design**, embedded systems, and microcontroller programming.
- Gained hands-on experience with hardware interfacing, **PCB design**, sensor integration, and communication protocols.
- Completed academic projects involving automation systems, robotics, and power electronics.
- Strong foundation in electronic device theory, signal processing, and instrumentation.

Projects

Commercially Deployed Projects

1. On-Board Diagnostics (OBD) Device for OHS Compliance – Telecom Sector

Jan 2025

- Designed and developed an edge-computing OBD system integrating **GPS, 6-DOF IMU (Accelerometer), dual-camera module, and 4G LTE SIM module**.
- Deployed **on-device ML inference** using optimized models for real-time driver behavior analysis and hazard detection.
- Implemented fault-tolerant firmware with power-efficient sleep/wake logic and real-time data synchronization to cloud infrastructure.

Technologies: C/C++, TensorFlow Lite, UART, CAN, MQTT, OTA, ARM Cortex-M, uBlox GNSS, ADXL345

2. Smart Three-Phase Energy Meter with Integrated Flow Sensor

March 2025

- Developed firmware for **industrial-grade 3-phase energy metering system** with **integrated solenoid-based flow control**.
- Enabled **real-time monitoring** and secure transmission of energy and flow data via **MQTT**.
- Included watchdog timers, persistent storage (EEPROM), and low-power modes for field reliability.

Technologies: Embedded C, Modbus RTU/TCP, SPI, ADC, STM32, IEC 62053 compliance

3. AI-Enabled Quadcopter Drone System – Telecom Inspection Use Case

April 2025

- Engineered a custom **quadrotor UAV platform** with integrated **NVIDIA Jetson Nano** for real-time object detection and telemetry analytics.
- Developed autonomous flight control system with sensor fusion from **barometer, GPS, IMU, and visual odometry**.
- Enabled aerial inspection, anomaly detection, and report generation for telecom infrastructure.

Technologies: ROS, PX4, MAVLink, Python, OpenCV, YOLOv5, RTSP Streaming, CAN, ESC/BEC integration

4. Fleet Management OBD System for Amazon Logistics

June 2025

- Designed a **dual-camera, telematics-enabled OBD system** for fleet vehicles, capturing and processing real-time driver telemetry and road conditions.
- Integrated **ML-based violation detection** (drowsiness, speeding, harsh braking, seat belt, smoking), and **event-driven cloud sync** with server.
- Implemented secure **OTA firmware updates** and **real-time alerting** system for compliance breaches.

Technologies: Embedded Linux, HTTP, TensorRT, GStreamer, RTOS, TLS, SD card logging, uClinux

5. Smart Body Measurement System for Tailoring – ML Edge Inference

July 2025

- Developed a **Raspberry Pi-based ML system** for real-time body measurement using camera and depth sensing, targeting virtual tailoring applications.
- Integrated with custom web dashboard for **virtual try-on and remote fitting services**.
- Optimized ML model pipeline for inference using **OpenVINO / TensorFlow Lite**, reducing latency and improving measurement accuracy.

Technologies: Python, Flask/Django, OpenCV, TensorFlow Lite, PiCam, USB depth sensor, SQLite, REST API

6. Research Project: Energy-Efficient Data Transmission Using LEACH Protocol and ML Inference

2023 - Ongoing

- Conducted **simulation-based analysis** of **LEACH (Low-Energy Adaptive Clustering Hierarchy)** protocol using **Cooja /Contiki-NG** to study energy-efficient data aggregation in wireless sensor networks (WSNs).
- Collected and preprocessed simulation telemetry for training an **ML model to predict optimal cluster head selection and data routing efficiency**.
- Deployed and tested the trained model on real hardware platforms including **ESP32** and **Raspberry Pi 4**, validating energy savings and transmission optimization in embedded edge environments.
- Evaluated system performance metrics such as **latency, power consumption, and packet delivery ratio**, aligning with goals for **adaptive and scalable WSN deployment in IoT ecosystems**.

Technologies: Cooja, Contiki-NG, Python, Scikit-learn/TensorFlow Lite, ESP-IDF, GPIO/I2C/SPI, Linux, MQTT, Edge AI

7. Ultrasonic Vibration Analysis for Machine Condition Monitoring

2024- Ongoing

- Use case: Detect bearing failures, cavitation, or motor misalignment.
- Machines emit ultrasonic signatures when defects start forming.
- High-frequency vibration acquisition. FFT + bandpass filtering for defect frequencies.
- Envelope detection for bearing fault harmonics.
- Predictive maintenance system that alerts before catastrophic failure.

Category	Protocols
Application Layer (IoT, Web & Messaging)	HTTP/HTTPS, SOAP, WebSocket, MQTT-SN, CoAP, AMQP, LwM2M, FOTA/OTA
Transport & Security	TCP, UDP, TLS/SSL
IoT Connectivity (LPWAN & Cellular)	NB-IoT, LTE-M, LoRa/LoRa WAN
Wireless Communication	BLE (4.0/5.0/5.3), Wi-Fi (802.11 X), ZigBee (IEEE 802.15.4), RFID/NFC
Wired Industrial Communication	UART (RS-232/RS-485/TTL), SPI, I2C, I2S, CAN/CAN-FD, LIN, Modbus RTU, Modbus TCP, Ethernet (IEEE 802.3), KNX
Research / Academic Protocol	LEACH

Software Tools 1. STM Cube IDE 2. Digital Signal Processing 3. NS-2 4. Keil IDE 5. Ki-Cad 6. Yocto Framework 7. Postman 8. Putty 9. MySQL 10. ESP IDF 11. UCLinux 12. Contiki/ Cooja	Hardware Analysis Tools 1. Oscilloscope 2. Signal Generator 3. Variable Power Supply 4. Logic Analyzer
Development Boards 1. ARM Cortex-MX 2. STM32 3. NVIDIA Jetson 4. Raspberry Pi 5 , 4, 3 5. AMB82-Mini 6. ESP32,32-S3, 8266 7. Arduino	Certifications 1. Ethical Hacker: IoT Attacks - Infosys 2. Ki-Cad PCB Designing 3. IoT Penetration Testing - Infosys 4. Android App Development 5. AI For India 2.0 - Geek Networks, IIT Madras 6. Python - Geek Networks, IIT Madras 7. Cloud Implementation using Azure IoT - Infosys 8. Industrial IoT 4.0-IIT Madras 9. ML Ops